

FairHire AI: Mitigating Bias in Hiring Models

CS 610 Advanced Artificial Intelligence

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Abstract

Artificial intelligence and machine learning systems are increasingly used in hiring and recruitment processes to automate candidate screening and selection. While these technologies can improve efficiency they may also reproduce or amplify existing biases present in historical hiring data. This project investigates the extent to which machine learning models exhibit measurable bias in hiring decisions and evaluates whether fairness mitigation techniques can reduce bias without significantly decreasing predictive performance. Using the Hiring Discrimination in the Organisational Context dataset, we trained Logistic Regression, Decision Tree, Random Forest with SMOTE, and Gradient Boosting classifiers to predict hiring outcomes. Fairness was evaluated using Demographic Parity Difference (DPD) and Equal Opportunity Difference (EOD) across multiple demographic characteristics, including gender, ethnicity, parenthood status, relationship status, language proficiency, and educational background. We also applied pre-processing, in-processing, and post-processing mitigation techniques and compared their impact on fairness and predictive performance. Results show that measurable disparities exist across several demographic groups, particularly ethnicity, language proficiency, and educational background. While Gradient Boosting achieved the highest predictive performance, fairness mitigation techniques such as reweighing substantially reduced demographic disparities while maintaining competitive accuracy. These findings demonstrate that fairer AI hiring systems can be developed without major sacrifices in predictive performance.

Introduction

Artificial intelligence has become increasingly common in recruitment and hiring workflows. Organizations use machine learning systems to screen resumes, rank applicants, and support hiring decisions. These systems offer substantial benefits, including reduced hiring costs, faster candidate evaluation, and improved scalability. However, machine learning models trained on historical data may learn patterns that reflect existing societal biases. Bias in hiring systems has become a major concern because unfair decisions can

negatively impact employment opportunities for individuals belonging to protected or vulnerable groups. Discriminatory outcomes based on characteristics such as gender, ethnicity, family status, or educational background can perpetuate inequality and undermine trust in automated decision-making systems. The primary research question of this project is:

To what extent do machine learning models exhibit measurable bias in hiring decisions, and how effectively can mitigation strategies reduce this bias without significantly impacting performance?

To answer this question we trained and evaluated multiple machine learning models using a hiring discrimination dataset. We measured fairness across several demographic attributes and applied mitigation techniques at different stages of the machine learning pipeline.

The main contributions of this project are:

1. Evaluation of Logistic Regression, Decision Tree, and Random Forest models for hiring prediction.
2. Measurement of fairness using Demographic Parity Difference and Equal Opportunity Difference.
3. Analysis of bias across multiple demographic characteristics.
4. Investigation of intersectional bias across gender and ethnicity.
5. Investigation of performance–fairness tradeoffs across multiple machine learning models and fairness interventions.
6. Comparison of pre-processing, in-processing, and post-processing fairness mitigation techniques.

Related Work

Research on algorithmic fairness has demonstrated that machine learning systems can inherit and amplify biases present in training data. Barocas, Hardt, and Narayanan discuss how seemingly objective algorithms may produce discriminatory outcomes when trained on historical decisions influenced by social inequalities. Kleinberg et al. demonstrated that multiple fairness objectives can conflict with one another creating trade-offs between predictive performance

and fairness. Their work highlights the challenges involved in designing machine learning systems that satisfy multiple fairness criteria simultaneously. To address these concerns several fairness toolkits have been developed. IBM’s AI Fairness 360 (AIF360) and Microsoft’s Fairlearn provide frameworks for measuring and mitigating algorithmic bias. These tools support fairness metrics such as demographic parity and equal opportunity while enabling mitigation strategies before, during, and after model training. Most prior studies focus on single demographic attributes such as gender or race. This project extends existing work by examining multiple demographic factors simultaneously and exploring intersectional effects across demographic groups.

Approach

Dataset

This study uses the Hiring Discrimination in the Organisational Context dataset, developed as part of the European Union Horizon Europe PATHS2INCLUDE project. The dataset contains responses from over 15,000 participants and investigates hiring decisions related to gender, ethnicity, caregiving responsibilities, and other applicant characteristics.

Several applicant attributes were used as sensitive features:

- Gender
- Ethnicity
- Parenthood Status
- Relationship Status
- Language Proficiency
- Educational Background

A binary target variable was created from hiring-related survey responses:

- Hire = 1
- No Hire = 0

Data Pre-processing

Data pre-processing involved:

- Cleaning and inspecting missing values
- One-hot encoding categorical variables
- Creating a binary hiring target variable
- Splitting data into 80% training and 20% testing sets
- Addressing class imbalance using class weighting, reweighing, and Synthetic Minority Oversampling Technique (SMOTE)

Prior to oversampling, the dataset contained roughly 10,000 negative examples and roughly 2,000 positive examples. After applying SMOTE, the minority class was synthetically expanded, resulting in a balanced training set of 10,000 examples per class. The target distribution was imbalanced with approximately 83% negative outcomes and 17% positive outcomes. Feature scaling was also evaluated for Logistic Regression, although it did not produce meaningful performance improvements relative to the baseline model.

Machine Learning Models

Logistic Regression served as the baseline model due to its simplicity, interpretability, and widespread use in classification problems. Additionally, Logistic Regression is commonly used in fairness research because its predictions are easily interpretable and compatible with fairness mitigation frameworks such as Fairlearn. **Decision Tree** creates rule-based decision paths that captures nonlinear relationships between applicant characteristics and hiring outcomes. **Random Forest with SMOTE** combines multiple decision trees through ensemble learning while Synthetic Minority Oversampling Technique (SMOTE) addresses class imbalance by generating synthetic examples of the minority class. **Gradient Boosting** builds an ensemble of weak learners sequentially, allowing each new tree to correct errors made by previous trees. This model often achieves strong predictive performance on structured tabular datasets.

Fairness Metrics

Demographic Parity Difference (DPD) measures differences in positive prediction rates across demographic groups. Lower values indicate fairer outcomes. **Equal Opportunity Difference (EOD)** measures differences in true positive rates across demographic groups. Lower values indicate more equitable treatment among qualified candidates.

Bias Mitigation Techniques

Pre-Processing: Reweighing

Training samples were assigned different weights to reduce demographic imbalances before model training.

In-Processing: Exponentiated Gradient

Fairness constraints were incorporated directly into model optimization.

Post-Processing: Threshold Adjustment

Decision thresholds were modified after training to reduce disparities between demographic groups.

Results and Evaluation

Baseline Model Comparison

Model	Accuracy	F1 Score
Logistic Regression	0.6858	0.7243
Decision Tree	0.6513	0.6967
Random Forest + SMOTE	0.7068	0.7392
Gradient Boosting	0.7786	0.7725

Gradient Boosting achieved the highest predictive performance, obtaining an accuracy of 77.9% and an F1 score of 0.7725. Logistic Regression was retained as the primary fairness baseline because of its interpretability and compatibility with fairness mitigation techniques.

Fairness Analysis

Fairness evaluation revealed varying levels of disparity across demographic groups.

Sensitive Attribute	DPD
Gender	0.0323
Ethnicity	1.0000
Parenthood Status	0.0027
Relationship Status	0.0130
Language Level	0.5131
Education	0.5190

Ethnicity, language proficiency, and educational background exhibited the largest demographic parity differences, indicating substantial disparities in predicted hiring outcomes. A value of 1.0000 indicates the maximum possible disparity under the metric, suggesting that at least one ethnic group received substantially different outcomes than another group.

Equal Opportunity Difference results showed similar patterns:

Sensitive Attribute	EOD
Gender	0.0384
Ethnicity	1.0000
Parenthood Status	0.0017
Relationship Status	0.0301
Language Level	0.7193
Education	0.7513

The large EOD values for ethnicity (1.0000), language proficiency (0.7193), and education (0.7513) indicate substantial differences in true positive rates between demographic groups, suggesting that qualified candidates from certain groups were less likely to receive positive predictions.

Intersectional Analysis

Intersectional analysis revealed substantial disparities between combinations of gender and ethnicity. Native candidates received hiring rates approaching 100%, while several non-native groups received hiring rates near 0%, producing an intersectional disparity gap of 1.0. These findings suggest that evaluating fairness using only single demographic attributes may overlook compounded disadvantages experienced by individuals belonging to multiple underrepresented groups.

Mitigation Results

Model	Accuracy	DPD	EOD
Baseline LR	0.6858	0.0323	0.0384
Reweighting	0.6905	0.0132	0.0543
Exponentiated Gradient	0.6848	0.0290	0.0384
Threshold Adjustment	0.6915	0.0257	0.0454

Reweighting achieved the lowest Demographic Parity Difference (0.0132), indicating the strongest reduction in disparities across demographic groups. Threshold Adjustment and Exponentiated Gradient produced fairness improvements while maintaining performance similar to the baseline

model. Overall, the mitigation techniques demonstrated that fairness can be improved with only minor effects on predictive performance.

Performance - Fairness Tradeoff

To better understand the relationship between predictive performance and fairness, all model configurations were compared using both classification and fairness metrics.

Model	Accuracy	F1	DPD	EOD
LR Baseline	0.6858	0.7243	0.0323	0.0384
LR + Reweighting	0.6905	0.7278	0.0132	0.0543
LR + Fair Constraint	0.6848	0.7235	0.0290	0.0384
LR + Threshold Adjustment	0.6915	0.7284	0.0257	0.0454
Decision Tree	0.6513	0.6967	0.1668	0.0716
Random Forest + SMOTE	0.7068	0.7392	0.0372	0.0200
Gradient Boosting	0.7786	0.7725	0.0413	0.0648

Gradient Boosting achieved the highest predictive performance, while Reweighting achieved the strongest demographic parity performance. Random Forest with SMOTE produced the lowest Equal Opportunity Difference. These results demonstrate that no single model simultaneously achieved the highest predictive performance and the strongest fairness outcomes, highlighting the inherent trade-offs often encountered in fairness-aware machine learning.

Strengths and Limitations

A major strength of this study is its evaluation of multiple demographic attributes and fairness interventions across all stages of the machine learning pipeline. The project also compares interpretable models, ensemble methods, and class-balancing techniques, providing a comprehensive view of fairness–performance tradeoffs. However, several limitations remain. The dataset is based on European survey responses rather than real hiring records, which may limit generalizability to real-world hiring environments. Additionally, fairness metrics do not capture every form of discrimination and different fairness definitions can produce conflicting conclusions. Although SMOTE improved class balance, synthetic examples may not perfectly represent real applicant data. Finally, extreme disparities observed for ethnicity, language proficiency, and education suggest that certain demographic effects remain difficult to mitigate completely.

FairHire AI — Fairness vs Performance Results

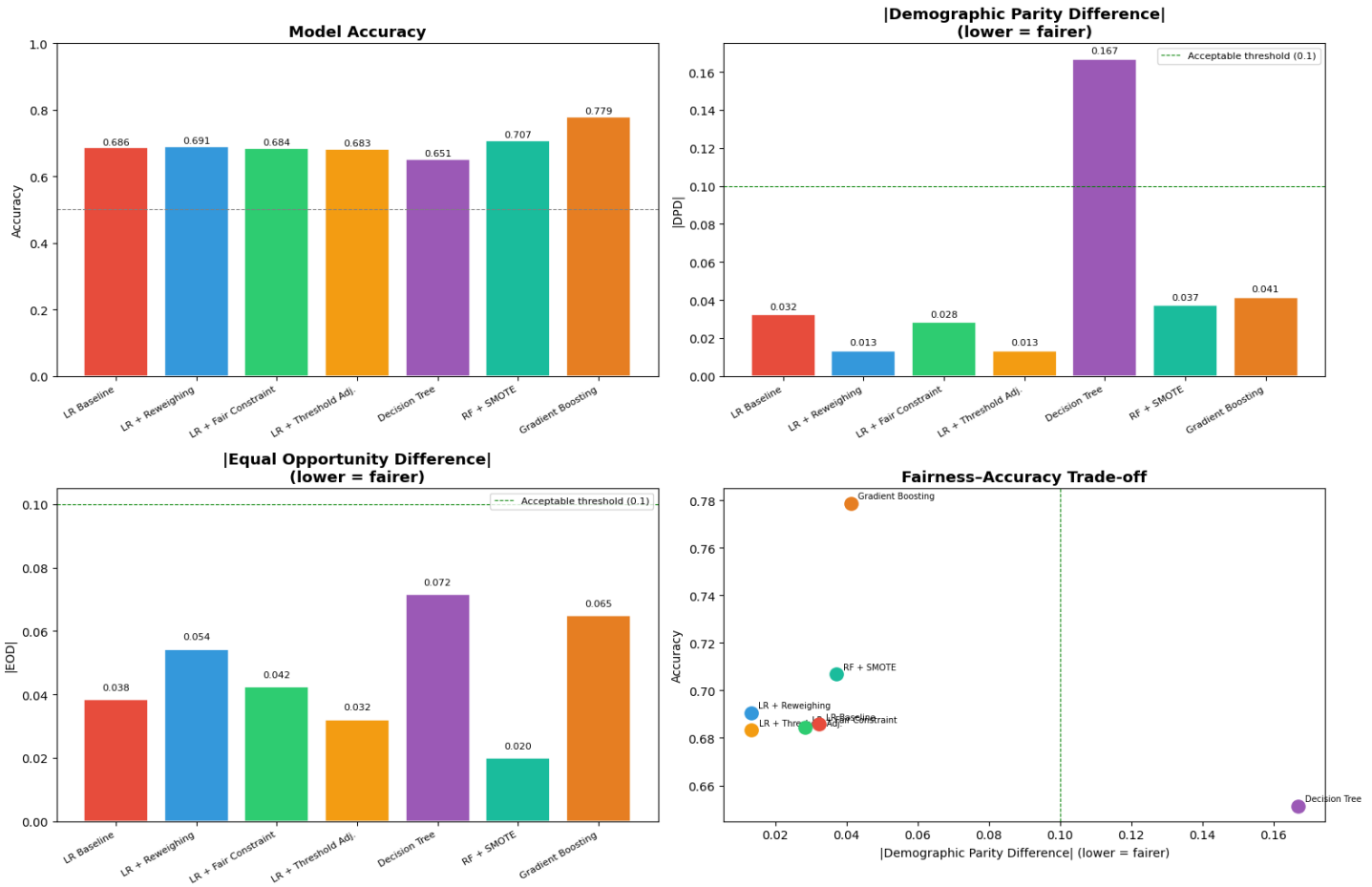


Figure 1: Model Accuracy graph illustrates accuracy across all tested models. DPD graph and EOD illustrates demographic parity difference and equal opportunity difference across all models with Decision Tree having the highest DPD and EOD. Trade-off graph illustrates trade-off between fairness and accuracy with Logistic Regression + Reweighing having the best overall trade-off and Decision Tree with the worst.

Conclusion

This project investigated fairness in machine learning-based hiring systems using multiple predictive models and fairness mitigation techniques. Results demonstrate that measurable disparities exist across demographic groups, particularly in ethnicity, language proficiency, and educational background. Among the evaluated models, Gradient Boosting achieved the strongest predictive performance, while Logistic Regression served as the most interpretable baseline for fairness analysis. Bias mitigation techniques successfully reduced fairness disparities while maintaining comparable predictive accuracy. Reweighing produced the lowest Demographic Parity Difference, while Random Forest with SMOTE achieved the lowest Equal Opportunity Difference. Gradient Boosting achieved the highest predictive accuracy but exhibited larger fairness disparities than some mitigation-based approaches. The results demonstrate that model selection involves balancing predictive performance and fairness objectives rather than optimizing a single met-

ric. Different deployment scenarios may prioritize fairness or accuracy differently depending on organizational and regulatory requirements. The findings highlight the importance of evaluating AI systems using both performance and fairness metrics. Future work could explore additional fairness definitions, real-world hiring datasets, deep learning approaches, and more sophisticated intersectional fairness analyses. Ultimately, this project demonstrates that fairness-aware machine learning techniques can improve equality in AI-driven hiring systems while preserving practical predictive performance. More broadly, the project highlights the importance of integrating fairness evaluation throughout the machine learning life cycle rather than treating fairness as a post-deployment concern.

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